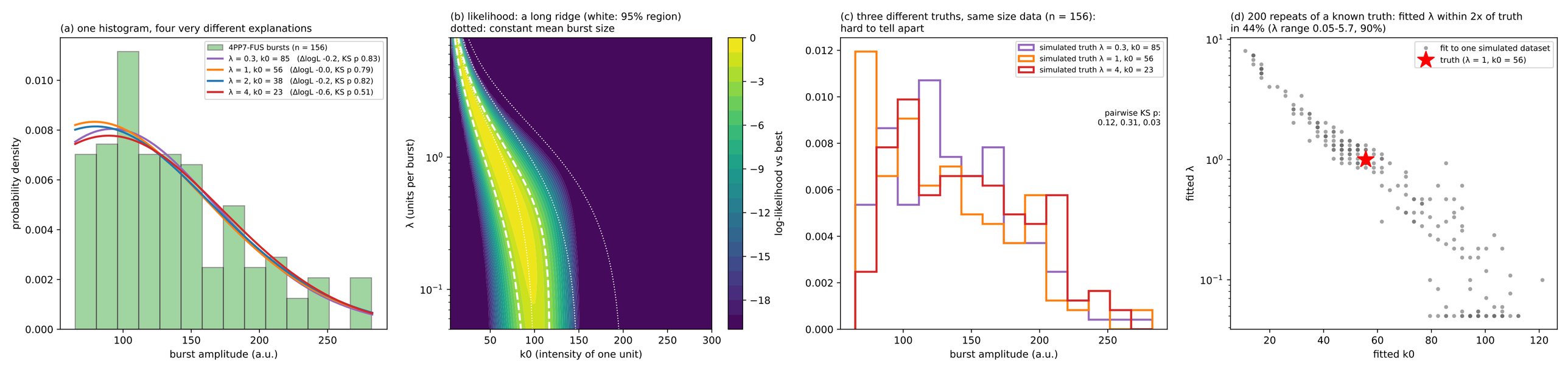

(a) one histogram, four very different explanations
4PP7-FUS bursts (n = 156)
λ = 0.3, k0 = 85 (ΔlogL -0.2, KS p 0.83)
λ = 1, k0 = 56 (ΔlogL -0.0, KS p 0.79)
λ = 2, k0 = 38 (ΔlogL -0.2, KS p 0.82)
λ = 4, k0 = 23 (ΔlogL -0.6, KS p 0.51)
probability density
burst amplitude (a.u.)
(b) likelihood: a long ridge (white: 95% region)
dotted: constant mean burst size
λ (units per burst)
k0 (intensity of one unit)
log-likelihood vs best
(c) three different truths, same size data (n = 156):
hard to tell apart
simulated truth λ = 0.3, k0 = 85
simulated truth λ = 1, k0 = 56
simulated truth λ = 4, k0 = 23
pairwise KS p:
0.12, 0.31, 0.03
burst amplitude (a.u.)
(d) 200 repeats of a known truth: fitted λ within 2x of truth
in 44% (λ range 0.05-5.7, 90%)
fit to one simulated dataset
truth (λ = 1, k0 = 56)
fitted λ
fitted k0